

Unit

2

Lesson

1



Innovators in Action.

Power Up Your Ideas!

Time Frame: [90 min]

Ready, Set, Learn!

You Need to Know

- Understand what a parallel circuit is and how electric current flows through multiple paths.
- Explain how voltage and current behave in a parallel circuit.
- Identify and use the breadboard to build simple circuits.
- Get introduced to Arduino and how it connects with components like LEDs and motors.
- Learn how to connect and control a servo motor using Arduino.
- Explore real-world uses of servo motors and how they respond to programmed signals.

Challenge questions of the lesson.

1. How can we design an automated system, like a smart bin, to solve a real-world problem effectively and hygienically, considering both its mechanical and electrical needs?
2. What are the fundamental differences between series and parallel circuits, and how does understanding these impact the design and reliability of multi-component smart devices?
3. How do various electronic components (sensors, motors, microcontrollers) work together as a "smart" system to sense, decide, and act, as seen in an automated dustbin?
4. Why is precision crucial in both physical circuit connections and digital 3D modelling, and how does it ensure the successful operation of our smart projects?

SEL Relation

- **Planning and Organization (Self-Management):** Students will practice planning the initial stages of their digital smart bin design in Tinkercad, thinking about where different compartments and parts will go. They'll also organize their ideas about how electronic components connect to make it work.
- **Problem-Solving (Self-Awareness & Responsible Decision-Making):** Students will encounter challenges in designing their bin in Tinkercad or figuring out how electricity makes sensors and motors function. They'll need to problem-solve how to make their digital design just right or understand tricky electrical connections.
- **Perseverance (Self-Management):** Working with digital design tools or trying to understand how electricity flows to make components work can sometimes be a bit tricky! Students will need to persevere through any difficulties they encounter, sticking with it until they figure it out.
- **Empathy (Social Awareness - Understanding others' perspectives):** Students will begin to think about how the layout of their smart bin or the clear design of their "Super-Guide" will affect how others easily understand and use the waste-sorting system, making it friendly for everyone!

Engage



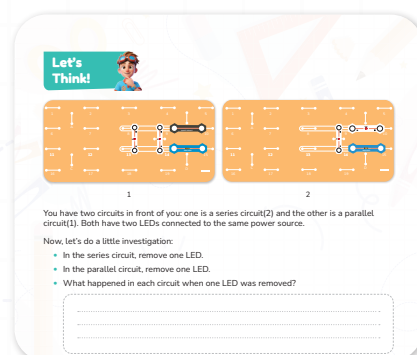
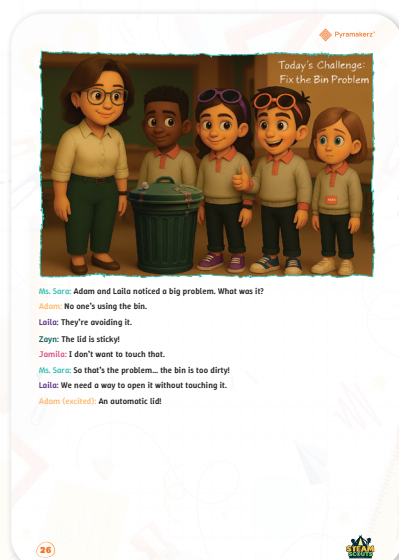
Teacher Role:

- **Facilitator:** Guide the discussion about the "dirty bin lid" problem, ensuring all students have a chance to share their thoughts and ideas about a touchless solution.
- **Questioner:** Ask open-ended questions to encourage critical thinking about how to automate the bin lid and introduce the concept of electricity as a power source.
- **Motivator:** Build excitement for learning about electricity, emphasizing how it powers smart solutions like an automatic bin lid.
- **Resource Organizer:** Prepare any materials needed for introducing electricity concepts (simple battery, wire, bulb for a quick demo if planned, or relevant visuals/videos).
- **Safety Monitor:** Ensure any introductory electricity demonstrations or future hands-on activities are conducted safely.

Expected From Students:

- **Active Listening:** Students will listen attentively to the dialogue and understand the challenge of the dirty/sticky bin lid.
- **Empathy:** Students will consider why peers might avoid the bin and recognize the importance of finding a hygienic solution.
- **Idea Generation:** Students will begin to think about different ways a bin lid could open without being touched.
- **Participation in Discussion:** Students will share their ideas and respond to suggestions made in the dialogue ("automatic lid," "electricity").
- **Problem-Solving:** Students will begin to think about what makes something "automatic" and why electricity is essential for smart devices.

Parts of the books included.



Story



- Teacher says (with a tone of observation, then excitement): "Today we're diving deeper into Adam and Laila's quest to solve the schoolyard trash problem! Remember how they noticed students weren't using the bins? Well, they've been investigating, and they've found the root of the problem – and they have a brilliant idea for a solution! Let's listen in as they discuss it..."

1. Engage with the Problem Dialogue.

- Purpose: This section uses the provided dialogue as a mini-play or shared reading to immerse students in the specific problem of the sticky bin lid, and to introduce the concept of an automatic lid and the subsequent need for electricity.
- Implementation:
 - Read Aloud with Roles: Assign roles (Mrs. Sara, Adam, Laila, Zayn, Jamila) and read the dialogue aloud with expression. Encourage different voices for each character to bring the scene to life.
 - Dialogue: Mrs. Sara: Adam and Laila noticed a big problem. What was it? Adam: No one's using the bin. Laila: They're avoiding it. Zayn: The lid is sticky! Jamila: I don't want to touch that. Mrs. Sara: So that's the problem... the bin is too dirty! Laila: We need a way to open it without touching it. Adam (excited): An automatic lid!
 - Pause and Discuss: After key lines or at the end of the dialogue, pause and ask guiding questions:
 1. "What's the main reason students are avoiding the bin, according to Zayn and Jamila?"
 2. "What's Laila's big idea to solve the dirty lid problem?"
 3. "What exciting concept does Adam introduce to make that happen?"

2. Ask a Think-Aloud Question

- Teacher asks: "An automatic lid! That sounds amazing. Have you ever seen something that moves all by itself, without anyone touching it? What do you think makes that possible? What powers it?"

3. Connect to the Activity

- Teacher explains: "Just like Adam and Laila realized their smart bin needs to open by itself, their automatic lid will need 'power' to work. Today, we're going to become 'Electricity Explorers' and figure out what electricity is and why it's so important for making things automatic and smart!"

Possible Strategies for Implementation:

Dialogue Reenactment & Problem Deep Dive

- How to implement:
 - Let students act out the dialogue that we just read, including the parts with

Mrs. Sara, Adam, Laila, Zayn, and Jamila. Encourage them to show the feelings of frustration about the sticky bin problem and excitement about the idea of an automatic lid.

- Pause precisely at Mrs. Sara's line, "Great! But to make it automatic, we need electricity. Who knows what electricity is?" This is the perfect moment to transition into a class discussion or a simple demonstration about electricity, launching the core learning about power for automation.
- Purpose: Engages students emotionally with the real-world problem (the dirty/sticky bin) and the proposed solution (an automatic lid), naturally leading to the introduction of electricity as the necessary power source.

Vocabulary and Concept Mapping

- How to implement:
 - Highlight and practice key words from the dialogue and discussion: avoiding, sticky, automatic, electricity, power.
 - Create a simple concept map on the board (or have students create their own):
 1. Problem: (Dirty/Sticky Bin Lid) -> Why? (Don't want to touch it / Avoiding it) ->.
 2. Solution Idea: (Automatic Lid) -> What makes it move? (Electricity/Power).
- Purpose: Reinforces new vocabulary and visually clarifies the logical progression from identifying a problem to proposing a smart solution and understanding its power source.

Let's Think!



Objectives:

Students will explore the difference between series and parallel circuits by observing what happens when one LED is removed from each type of circuit. This hands-on investigation supports their.

Activity Instructions:

1. Show students the two circuits in the image:
 - Circuit 1: Parallel.
 - Circuit 2: Series.
2. Explain that both circuits have two LEDs connected to the same power source.
3. Ask students to remove one LED from each circuit, starting with the series circuit, then the parallel one.
4. Instruct students to observe carefully and write down what happened in each circuit when the LED was removed.

Possible Strategies for Implementation:

- Hands-On Lab: If using real components, support students in carefully removing one LED at a time and observing the outcome.
- Virtual Circuit Testing: If using Tinkercad, guide students to simulate the LED removal and watch the results live.
- Group Work: Pair students to help each other during the test. Let one remove the LED while the other records observations.
- Whole-Class Discussion: After everyone tests the circuits, lead a quick discussion asking:
 - "Which circuit kept working?"
 - "Why did only one LED go off in the parallel circuit?"

Explorer's Toolkit



Teacher Role:

- Introducer: Let's watch this video about a smart dustbin! It's not just a regular bin; it sorts wet and dry waste all by itself!
- Facilitator: Guide students to pay close attention to the electric parts in the video. What are they? Where are they?
- Questioner: Prompt students to think about how these electric parts work together. How does the bin know when to open and close? What makes it "smart"?
- Resource Manager: Make sure students have access to the video link: http://www.youtube.com/watch?v=qOer_loJg4g



- (play just first 0.33 seconds to observe how the smart bin works, you can repeat this part to make sure all the students take their notes).
- Observer: Observe how students identify the different parts and explain their function.

Expected From Students:

- Active Viewing: Students will watch the video carefully, focusing on the electric parts of the smart dustbin.
- Identification: Students will identify the electric parts used in the project: Servo SG90, Ultrasonic Sensor, Arduino Uno, Soil Moisture Sensor, and the 16x2 LCD with i2c interface.
- Explanation: Students will explain how these parts work together to automatically open and close the dustbin. The soil moisture sensor detects if the garbage is wet or dry.
- Application: Students will relate these concepts to their own smart bin designs.

Parts of the books included



DIY Lab

This is our smart project—a dustbin that can automatically separate wet and dry waste on its own.

As you watch, look carefully at the electrical components like sensors, wires, and motors. See how they all work together to make the bin open and close by itself.

https://www.youtube.com/watch?v=qOer_loJg4g



Component Name	Image	I Know This!
Battery		☐
LED (Light Emitting Diode)		☐
Resistor		☐
Switch (Toggle)		☐
Wire		☐
Breadboard		☐
Arduino Board		☐
Servo Motor		☐

Figure 16: Smart Dustbin Components.

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DIY Lab

Objectives:

Students will investigate how different electronic components (like sensors and motors) work together to make a smart dustbin automate the sorting of wet and dry waste. By carefully watching the video, students will apply observation and critical thinking to understand the principles of automation, how technology solves problems, and how individual parts create a functional "smart" system.

Materials Needed:

- Laptop with internet access

Video link: https://www.youtube.com/watch?v=qOer_loJg4g

- Pencil or pen

Activity Instructions.

Part 1: Discovering the Electric Parts.

- Watch for the Parts: Watch the video once, from start to finish. As you watch, keep an eye out for all the different electric parts the inventor uses.
- Identify and List: check on the electric parts you see (like those little circuit boards, wires, or things that move).
- Think About Their Job: For each electric part you identify, quickly jot down what you think its job might be in the dustbin. Does it see things? Does it move things? Does it think?

Part 2: How Do They Work Together?

- Re-Watch with a Mission: Now, watch the video again, but this time, really focus on how these parts connect and what happens when they work together.



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- Follow the Flow:
 - The "Eyes": How does the dustbin "see" if there's trash coming? What electric part does that?
 - The "Brain": What part seems to be the "brain" that makes all the decisions?
 - The "Action": What electric part actually makes the dustbin lid open and close?
 - The "Decision Maker" (Wet or Dry?): How does the dustbin decide if the trash is wet or dry? What special electric part helps with that?
- Connect the Dots: In your notes, try to describe how these different parts talk to each other to make the dustbin open automatically for wet or dry waste.

Hands-On Activity: Blink an LED with Arduino.

Activity Instructions for Teachers.

1. Play the video [<https://www.youtube.com/watch?v=NREtn59WPho>] for the class to learn about breadboard.



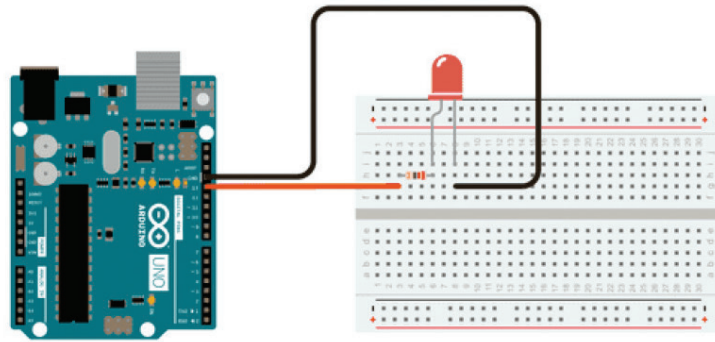
2. Ask students to open Tinkercad Circuits and replicate the same blinking LED circuit shown in the book.
3. Support students in:
 - Dragging an Arduino and LED onto the canvas.
 - Writing the Arduino code that turns the LED on and off repeatedly.
 - Uploading and running the simulation.
4. Once the LED blinks correctly, encourage students to try changing the delay values to observe the effect on the blink speed.

Materials Needed in Tinkercad:

- Arduino board
- LED
- Resistor (220Ω)
- Breadboard
- Arduino IDE

Step-by-Step Instructions:

- Connect the LED to the Arduino:
 - Short leg (negative) → GND (ground) on Arduino.
 - Long leg (positive) → Digital Pin 13 (via a resistor).
- Open the code section and type this code:



```
void setup () {  
  pinMode (13, OUTPUT);  
}  
  
void loop () {  
  digitalWrite (13, HIGH); // Turn LED on  
  delay (1000);           // Wait 1 second  
  digitalWrite (13, LOW); // Turn LED off  
  delay (1000);           // Wait 1 second  
}
```

Start the simulation.

Observe the LED blinking every 1 second.

Experiment:

- Change the delay values (e.g., delay (500); for a faster blink).
- Connect the LED to a different pin and update the code (pinMode (9, OUTPUT) ;).

Explain



Teacher Role:

- Spark curiosity by introducing Arduino as a real-world tool used in smart technologies.
- Briefly explain what students will see in the video and how it connects to their project.
- Pause the video at key moments to ask guiding questions (e.g., “What can Arduino control in real life?”).
- Help students make connections between the video and their own smart bin project.
- Encourage discussion after the video about how Arduino acts as the brain of their designs.

Expected From Students:

- Watch the video actively and take note of what Arduino is and what it can do.
- Identify examples of how Arduino is used to control devices or perform tasks.
- Participate in post-video discussion or respond to prompts like:
 - “What did you learn about Arduino?”
 - “How might you use it in your smart bin?”
- Begin thinking about how Arduino connects with sensors and outputs in their own projects.

Parts of the books included.



Screen Lessons

Imagine a tiny, super-smart computer that you can program to do almost anything! That's what an Arduino is.

So, let's explore the world of Arduino.

https://www.youtube.com/watch?v=1ENiVwk8idM&ab_channel=Fireship



Screen Lessons

Objective:

Students will understand what an Arduino is, how it functions as a programmable microcontroller, and how it can be used to control components like LEDs, motors, and sensors in real-world smart devices.

Video link: https://www.youtube.com/watch?v=1ENiVwk8idM&ab_channel=Fireship



Possible Strategies for implementation:

1. Before Watching: Setting the Stage for Smartness

Strategy: What Makes Something Smart?

Before starting the video, ask students:

“What do you think makes an object ‘smart’? What should it be able to do?”

List their responses on the board—such as: “It should sense things,” “It should respond,” “It should think or decide something.”

Explain that they will now see how Arduino helps us make everyday objects smart by giving them a brain!

2. During Watching: Active Engagement with Electronics

Strategy: Arduino Parts Checklist

Before the video, hand out a short checklist for students to complete while watching. Ask them to look for:

- What does the Arduino board look like?
- Can you name or describe any parts you see?
- What does the code do on the board?



(Video link: <https://g.co/gemini/share/5142ce3c045b>)

3. After Watching: Reflection and Application

Strategy 1: Smart Bin Brainstorm.

Ask: “Now that we know what Arduino can do, how could it be used in our smart bin?”

Guide students to brainstorm how Arduino could act as the “brain” of the bin—for example, reading a sensor and sending signals to show a message or light up an LED.

Strategy 2: Design a Smart Action.

In pairs or groups, have students design a smart action their dustbin could do using Arduino.

Examples:

- “If the bin is full, an LED lights up.”
- “The bin shows ‘Full’ or ‘Empty’ on an LCD screen.”

Students should describe what sensor might send the signal and what Arduino will do with it.

Strategy 3: Sketch a Smart System

Have students draw a basic diagram showing how the smart bin works.

They should include the Arduino, sensors, and outputs (like lights or screens), labeling each part and using arrows to show how information moves from sensor → Arduino → action.

Elaborate and Evaluate



Teacher Role:

- **Introducer & Guide:** Set the stage for our servo motor test! Remind everyone of our goal: to get a servo moving by giving it a name, connecting it, and writing some simple code. Make it clear this is how we bring electric projects to life!
- **Concept Clarifier:** Help students truly understand why each part matters. What do wires do? Why do we use a servo here? And who's the real "brain" behind the magic? Clarify any tricky vocabulary.
- **Hands-On Facilitator (or Conceptual Guide):** Whether we're actually connecting a servo or just thinking through the steps, guide students. Show them (or describe) how to make that simple connection and introduce the idea of "code" that makes it all happen.
- **Problem-Solver Supporter:** As students tackle the "What do wires do?", "Why a servo?", and "Who's the brain?" questions, gently nudge their thinking. Help them

recall previous videos or brainstorm answers.

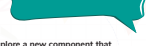
- Observer & Cheerleader: Watch as students light up with understanding! Observe how they reason through the questions and connect the dots between the parts and the power.

Expected From Students:

- Active Engagement: Students will actively participate in learning how to connect and conceptually "code" a servo motor, ready to bring it to life!
- Question Masters: Students will think hard about and answer the multiple-choice questions:
 1. What do wires do in an electric project?
 2. Why do we use a servo motor in a project?
 3. Which component acts as the brain of the system?
- Reasoning Power: They won't just pick an answer; they'll be ready to explain why they chose it, showing their awesome understanding of wires, servos, and the mighty Arduino!
- Connecting the Dots: Students will connect their new knowledge of servos, wires, and the Arduino to understand how these pieces work together to make automated projects, like our smart dustbin, really happen.

Parts of the books included.





Today we're going to explore a new component that helps us keep our circuits neat and organized. It's called a breadboard—let's find out how it works!

Click the link to learn more about what a breadboard is, then try building the LED circuit yourself in Tinkercad:

<https://www.youtube.com/watch?v=nREtE59YWPg>



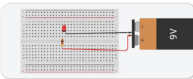


Figure 17: LED Circuit on a Breadboard.

Screen Lessons

Imagine a tiny, super-smart computer that you can program to do almost anything! That's what an Arduino is.

So, let's explore the world of Arduino.

https://www.youtube.com/watch?v=1ENYw8DiM5ab_channel=Firefish



Assessment



Apprentice

1. What do wires do in an electric project?

- Hold things together
- Send electricity
- Make noise

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2. Why do we use a servo motor in a project?

- To make lights blink.
- To move parts to a specific position.
- To store information.

3. Which component acts as the brain of the system?



□ Arduino



□ Bread board



□ Ultra Sonic

Let's Wrap Up What We Have Learned!

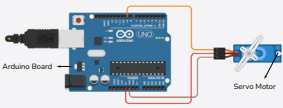
- Arduino is a small, smart computer you can program to control things like lights, motors, and sensors. It helps bring your ideas to life by making projects work automatically!
- A servo motor is a small motor that moves to the exact spot you tell it to. It helps robots and machines move parts just right.





Builder

Connect a servo motor to an Arduino.



Arduino Board

Servo Motor

Figure 18: Arduino circuit with servo motor.






Wire Color	Connect To (Arduino Pin)	What it Does
● Red	5V	This gives power to the servo. It's like 'Twirly's' energy!
● Brown or ● Black	GND	This is the ground. It keeps the power safe and flowing right.
● Orange or ● Yellow	Pin 9	This is the signal wire. It tells 'Twirly' where to move!



Mastermind

Put the mixed-up code lines in the correct order, then upload it to Tinkercad to make your servo circuit from before work properly.



Code Reorder Puzzle: Make a Servo Rotate!

Arrange the code lines so the servo motor rotates from 0 to 90 to 180 degrees. Enter the letters in order (e.g. A,C,B,3).

Scrambled Code Lines:

L `writeMicroseconds(180); // Move to 180 degrees`

K `delay(1000);`

J

A `writeMicroseconds(90);`

J `writeMicroseconds(180); // Move to 180 degrees`

N `3`

I `delay(1000);`

Your Order:

1 2 3 4 5 6 7 8 9 10 11 12 M N

Check Code



Apprentice

Objective:

Students will demonstrate their understanding of basic electronic components like wires, servo motors, and the Arduino, by answering questions about their functions in a project. This activity helps build essential technology vocabulary, develops critical thinking about how parts work together, and reinforces foundational knowledge for creating smart systems.

Correct Answers:

Apprentice

1. What do wires do in an electric project?

- A. Hold things together
- B. Send electricity
- C. Make noise

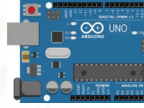
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2. Why do we use a servo motor in a project?

- A. To make lights blink.
- B. To move parts to a specific position.
- C. To store information.

3. Which component acts as the brain of the system?



Arduino



Bread board



Ultra Sonic

Possible Strategies for Implementation:

1. Recall Your Knowledge (Brain Power!):

- Strategy: Before looking at the answer choices, have students think back to our videos about Arduino and Servo motors, and our discussions about how wires work. What do they already know about each component?
- Why it helps: This strategy encourages them to rely on their own learned knowledge first, strengthening their memory and understanding.

2. Read Carefully (Be a Detective!):

- Strategy: Encourage students to read each question and all the answer choices very carefully. Sometimes, an answer might sound right, but one is even more correct or specific to what we learned.
- Why it helps: This helps students avoid tricky distractions and find the best fit for the question.

3. Think About Their "Job" (What's Their Superpower?):

- Strategy: For each question, have students imagine the component (like a wire or a servo motor) and think about its main "job" or "superpower" in an electric project. What does it do?
- Why it helps: Connecting the component to its key function will lead them to the correct answer, especially for questions about "why" we use it or "what" it does. For the "brain" question, they should remember what we called the main controller.



Builder

Objective:

Students will learn how to connect and send simple instructions (code) to a servo motor, making it move to specific positions. This activity will help them understand the fundamental steps of controlling an electronic component, building skills in practical electronics, following instructions, and seeing code in action.

Builder

Connect a servo motor to an Arduino.

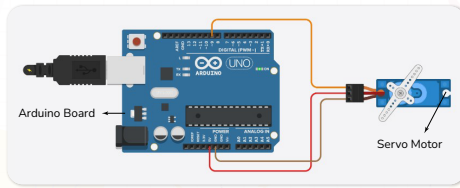


Figure 18: Arduino circuit with servo motor.

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Pyramakerz®

Servo Wires Colour Guide:

Wire Color	Connect To (Arduino Pin)	What It Does
Red	5V	This gives power to the servo. It's like Twirly's energy!
Brown or Black	GND	This is the ground. It keeps the power safe and flowing right.
Orange or Yellow	Pin 9	This is the signal wire. It tells Twirly where to move!

Materials Needed:

- Arduino board (Arduino Uno).
- Servo motor
- Jumper wires (male-to-male).
- For Conceptual/Simulated (If actual materials aren't available):
 - Computer/tablet with internet access for a simulator (Tinkercad Circuits).
 - Diagrams showing servo connections to Arduino.

Activity Instructions:

- Introduce the Mission: Explain to students: "Today is super exciting! We're going to make a servo motor actually move! Remember how a servo helps our smart dustbin lid open just right? Well, today, we'll learn the steps to make it move ourselves!"
- Meet Our Servo: Hold up a servo motor (or show a clear picture/animation). "This little gadget is our servo motor! It has wires, and its job is to move to exact spots. We're going to learn how to tell it exactly where to go." (mention that it has long range of angles so they will try until they get the exact angle)

Step 1: Making the Connection (The Wires!).

- For Actual Hands-on: Guide students to identify the three wires on the servo (power, ground, signal). Show them how to connect them correctly to the Arduino and breadboard. Emphasize matching colors/functions.
- For Conceptual/Simulated: Project or hand out a clear diagram of the servo motor connected to an Arduino. Discuss which wire goes where and why ("power needs power, ground needs ground, and signal is where the 'instructions' come from!").

Step 2: Giving It a Name (and Writing the Code!).

- For Actual Hands-on: Guide students to open the Arduino IDE. Show them how to include the Servo library and how to "name" their servo in the code (Servo myServo;). Explain the basic setup () and loop () functions. (just explanation cause that will be their guide in mastermind).
- For Conceptual/Simulated: Show a very simple code snippet on screen. Explain that code is like giving instructions to the Arduino. Point out where we "name" the servo and where we'd tell it to move (myServo.write(0); for 0 degrees, myServo.write(90); for 90 degrees).
- Note: just explain that code to them but do not let them write it, upload it only.


```

#include <Servo.h>
Servo myServo; // Create servo object
void setup () {
  myServo.attach(9); // Attach servo signal
  wire to pin 9
}
void loop () {
  // Move from 0 to 180 degrees
  for (int pos = 0; pos <= 180; pos += 1) {
    myServo.write(pos);
    delay (15); // Small delay for smooth
    movement
  }
  // Move from 180 to 0 degrees
  for (int pos = 180; pos >= 0; pos -= 1) {
    myServo.write(pos);
    delay (15);
  }
}

```

- Encourage Observation & Reflection:
 - As they connect, code:
 1. "Did your servo move the way you expected?"
 2. "What did the numbers in the code (like 0 or 90) make the servo do?"
 3. "How is this like telling our smart dustbin to open its lid just a little, or all the way?"
- Facilitate Sharing:
 - Pair students or have a quick group share. Ask:
 1. "What was the coolest thing you learned about making the servo move?"
 2. "How does understanding this help us think about building our smart dustbin even better?"



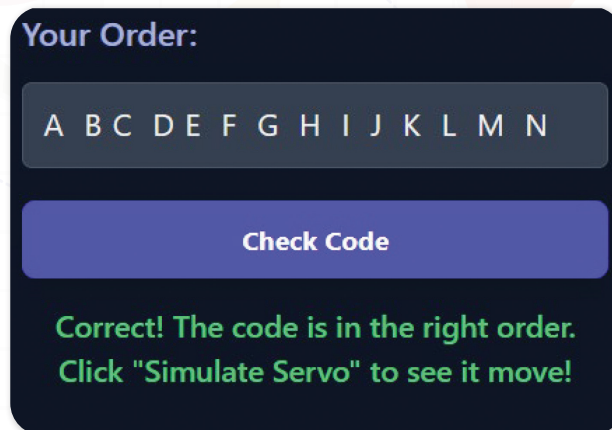
Mastermind

Objective:

Students will apply their understanding of basic programming logic by arranging scrambled code lines to make a servo motor rotate to specific angles. This activity will

help them practice critical thinking, recognize fundamental code structure, and develop sequential problem-solving skills essential for controlling electronic components.

Answer:



Activity Instructions.

1. **Introduce the Mission:** "You've learned about how clever 'brains' (like Arduino) can control 'muscles' (like Servos)! Now, it's time to put on your coder hats and tackle a puzzle. Your mission is to unscramble some mixed-up code lines to make a servo motor move in a specific way!"
2. **Clarify the Goal:** "When you finish arranging the code, here's what should happen: The servo motor should rotate from 0 degrees, then move to 90 degrees, and finally to 180 degrees, with a short pause after each move. You need to:
 - Figure out the correct sequence of the code lines.
 - Make sure the code makes sense from top to bottom to achieve our goal."
3. **Emphasize Logic:** "This isn't about guessing! It's like solving a puzzle where each line has a special place. Think about what needs to happen first to set up the servo, and then what sequence of movements and pauses should repeat for our servo to 'dance' just right."

Possible Strategies for Implementation:

1. **"Code Storyteller: Read All Lines & Predict the Goal!"**
 - **How to Implement:** Before moving any lines, have students read all the scrambled code lines from beginning to end. Then, specifically ask: "Based on these lines, what do you think this code is trying to make the servo do?" Connect this directly to the stated goal of rotating the servo to 0, 90, and 180 degrees.
 - **Why it Helps:** This gives them a general idea of the code's purpose and helps them spot common programming "phrases" or instructions related to servos and timing (delay).
2. **"The Main Acts: setup() and loop()!"**
 - **How to Implement:** Guide students to specifically look for void setup() and void loop() (referencing the common Arduino structure). Remind them that setup() usually happens once (like getting ready to use the servo!), and loop() happens over and over again (the main performance where the servo moves repeatedly!).

- Why it Helps: These two functions are the foundational structure of Arduino code; finding them helps structure the entire code puzzle.

3. "Casting the Character: Servo Declaration & Attachment!"

- How to Implement: Encourage students to look for lines where the servo is "named" or set up, specifically `#include <Servo.h>`, `Servo myServo;`, and `myServo.attach()`. Explain that these lines tell the Arduino that we're going to use a servo and where it's connected.
- Why it Helps: Getting these initial setup and definition lines in place early ensures the program knows what component it's working with before it tries to use it.

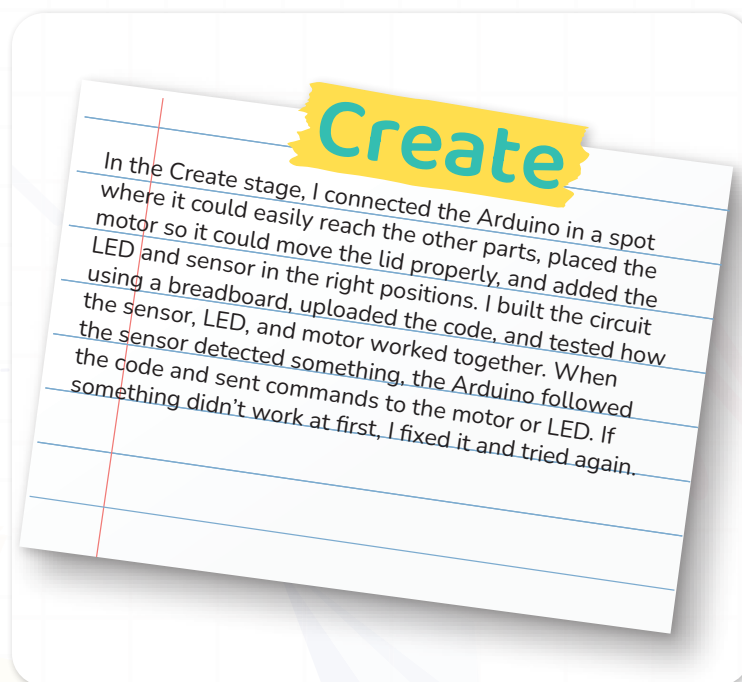
4. "The Servo's Dance Steps: `write()` and `delay()`!"

- How to Implement: Guide students to specifically look for `myServo.write()` commands (which tell the servo to go to a certain angle) and `delay()` commands (which make the program pause). Ask: "What's the order of angles the servo needs to go to (0, 90, 180)? Where do we put the pauses so we can see each move?"
- Why it Helps: This focuses on the core sequential logic of the servo's movement and timing, which directly relates to the puzzle's outcome.

5. "Mental Rehearsal: Play the Code in Your Head!"

- How to Implement: Once students have an order they think is correct, have them "read" the code out loud or in their head, imagining the sequence of events. "First, the servo gets ready. Then, it moves to 0, pauses. Then it moves to 90, pauses. Then it moves to 180, pauses. Does this repeat?"
- Why it Helps: This self-checking strategy helps them find any mistakes in their logical flow before actually testing the code.

Don't forget to go back to the EDP page and add these to creating part.





Now I Can

Objective:

Students will reflect on and self-assess their learning and practical skills gained throughout our smart dustbin design project. By reviewing and checking off "I Can" statements, they'll demonstrate their understanding of key concepts.

Activity Instructions

1. Read each statement aloud one by one and pause to let students reflect.
2. Ask students to think of an example or recall an activity they did that matches each skill before checking the box.
3. For any statements they don't check, reassure them it's okay and encourage them to ask for help or revisit that topic.
4. Ask students to write a short sentence or draw a symbol next to each checked box to show how they demonstrated that skill (e.g., sketch a breadboard, write "I made a blinking LED").
5. Encourage peer discussion or sharing their "Now I Can" reflections in pairs or small groups.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.
 - (Students are constructing a physical model of their smart bin's sensor-motor system using an Arduino, IR sensor, and servo motor. They could also draw a circuit diagram or a conceptual sketch illustrating how the IR sensor's "vision" leads to the servo's movement, demonstrating how the arrangement and function of these electronic components help solve the problem of an automatic lid.).
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.
 - (As students test their IR sensor and servo circuit, they are implicitly analyzing data by observing how the system performs when an object is detected versus when it's not. They might also compare how the servo responds to different code values or how the sensor performs under different lighting conditions, noting the strengths and weaknesses of the system's reaction based on the sensor's input.).

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings.
 - (The physical circuit build itself serves as a visual display of their design ideas. Students can also create labeled circuit diagrams or flowcharts illustrating their code's sequence to clarify how the IR sensor triggers the servo, enhancing their

explanation of their smart bin's automatic function.).

3. Common Core State Standards (CCSS) - Mathematics (Geometry):

- **1.G.A.2:** Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) or three-dimensional shapes (cubes, right rectangular prisms, right circular cones, and right circular cylinders) to create a composite shape, and compose new shapes from the composite shapes.
 - (Students are using their understanding of 3D shapes to envision and potentially plan the placement of components like the IR sensor, servo motor, and Arduino within or on the
 - composite shape of their smart trash bin. They compose these individual components into a functional arrangement, much like composing shapes to create a larger structure.